

Aayush Gupta

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Education

Chhatrapati Shahu Ji Maharaj University, Kanpur
Bachelor Of Science (Mathematics & Computer Science)

July 2024 – Present
CGPA: 7.54 [up to semester 4]

Technical Skills

Systems & Scientific Computing: C, Python/C API, NumPy C-API, CMake, scikit-build-core, cibuildwheel

Languages: Python, JavaScript, C

Web & API: React, Node.js, Express, REST APIs

Databases: MongoDB, MySQL, Firebase, Redis

DevOps & Cloud: GitHub Actions, Docker, Docker Compose, AWS (EC2, S3), Google Cloud (Vertex AI)

Tools: Git & GitHub, VS Code, Linux (Ubuntu)

Work Experience

Scientific Software Open Source Engineering Contributor -
[Google Summer Of Code 2026](#)

April 2026– August 2026

QC-Devs (Quantum Chemistry) - Remote

[Final Report](#)

- Migrated the GBasis build system from setuptools to **scikit-build-core** with **CMake FetchContent**, replacing manual libcint compilation with a single “pip install” command; shipped **cross-platform wheels (Linux/macOS/Windows, x86/ARM)** via **cibuildwheel** and **GitHub Actions CI/CD**.
- Designed a C shell-loop integration engine using the **Python/C API** and **NumPy C-API** with **token-pasting macros**, replacing slow per-call ctypes bindings for one- and two-electron integrals.
- Delivered measured **150x–353x speedups** over the baseline Python implementation (e.g., **353x on overlap integrals, cc-pVDZ basis, H₂O**), **benchmarked on Apple M5 hardware** with reproducible before/after methodology.
- Implemented platform-aware shared library loading (.so/.dylib/.dll) with automatic x86 vs. ARM architecture detection at build time, **fixing libcint v6 naming inconsistencies across platforms**.

Built a **1,146-test suite** (730 libcint-specific), validating gradient and 3-center integrals against reference implementations within **1e-10 numerical**

Scientific Software Open Source Engineering contributor
[Winter Of Code 5.0 program](#)

Jan 2026 – March 2026

QC-Devs (Quantum Chemistry) - Remote

- Reimplemented **2-electron repulsion integral** engine using **Obara-Saika** and **Head-Gordon-Pople** (OS+HGP) algorithm with **VRR**, **ETR**, and **HRR recursion** stages across 5 core modules.
- Implemented **Boys function** with 4 variants (standard, erf, erfc, Yukawa) using **vectorized NumPy** with mpmath fallback for arbitrary angular momentum orders
- Built **Schwarz screening** module reducing integral computation from **O(n⁴) to O(n²)** shell-pair bounds with caching, skipping negligible shell quartets.
- Validated against **PySCF** reference implementation for s, p, d orbitals — **364 tests passed**, 0 failed, within **1e-10 tolerance**.
- Closed 3 open issues ([#221](#), [#216](#), [#208](#)) and shipped work across **9 merged PRs** in 9 weeks.

Open Source Contributions

Smart Resume Reviewer (10 Pull Requests) | Python, Streamlit **Winter of Code Social | Jan 2026**

- Implemented batch upload feature enabling simultaneous analysis of 2–10 resumes with side-by-side comparison table, color-coded ATS scores, score distribution charts, and CSV/PDF export with timestamped filenames.
- Built resume preview functionality using PyMuPDF to render PDF first-page as image and display file metadata before analysis, reducing user errors and improving UX confidence.

AgriTech (2 Pull Requests) | Documentation

Social Winter of Code (SWOC) | Jan 2026

- Improved project README by adding Table of Contents, API key setup guide (Gemini/Firebase), Production vs Development notes, Issue Reporting section, FAQ, and License — enhancing onboarding clarity for new contributors.
- Resolved documentation gaps that were causing confusion for users setting up the project locally, improving contributor experience for a repository with 103+ stars and 242+ forks.

Projects

Bowsensei - AI Archery Coaching System (Kaggle) | Python, Vertex AI, Google Cloud **Nov2025 - Jan 2026**

- Built an AI-powered archery coaching system as capstone project for Google × Kaggle Agentic AI Program, applying agent-based AI concepts including MCP, context engineering, and production readiness.
- Deployed on Google Cloud using Vertex AI; project appreciated by a Google engineer for real-world application and execution.
- Authored technical articles on Agentic AI as part of the Dev.to writing challenge.

Achievements & Extra-Curricular

- Shortlisted in **KPMG Malta Full Stack Azure Hiring Challenge** (HackerEarth) as Team Leader — proposed a healthcare platform for virtual check-ups; idea awarded an incentive of ₹3,750 among top submissions.
- Led a 5-member team in **Smart India Hackathon** — developed a software solution for speech sound disorder intervention under the Smart Education theme.
- Built an **Anti-Deepfake detection prototype** during Google DeepMind × Kaggle "Vibe Code with Gemini" hackathon using LLM-based detection workflows.
- Selected for **Winter School on Cognitive Modelling — IIT Mandi**, focused on computational thinking, human cognition modelling, and interdisciplinary problem-solving frameworks.
- Completed **Google × Kaggle Agentic AI Capstone** with deployment on Vertex AI.